

$$1. SP = [x := x + 2, x \geq 2]$$

$$\exists x_0, x_0 = 0 \quad x = x_0 + 2$$

$$\boxed{x \geq 2}$$

$$2. SP = [y := y + x, x \geq 1 \wedge y \geq 0]$$

$$\exists y_0, x \geq 1 \wedge y_0 \geq 0 \wedge y = y_0 + x$$

$$\boxed{x \geq 1 \wedge y_0 \geq 0 \wedge y = y_0 + x}$$

$$3. SP = [x := x - 3, x \geq 0]$$

$$\exists x_0, x_0 \geq 0 \wedge x = x_0 - 3$$

$$\boxed{x \geq 3}$$

$$4. SP = [u := u + 10, u < v]$$

$$\exists u, u_0 < v \wedge u = u_0 + 10$$

~~u - 10 < v~~

$$\boxed{u - 10 < v}$$

$$5. SP = [u := u + 3, u < v \wedge w = 0]$$

$$\exists u_0, u_0 < v \wedge w = 0 \wedge u = u_0 + 3$$

$$u = v + 3, u < v \quad \times$$

$$6. SP = [x := 42, x = y + 1]$$

$$\exists x_0, x_0 = y + 1 \wedge x = 42$$

$$y = 41$$

$$7. SP = [x := y; y := x + 1, y = 5]$$

$$y = x + 1, y_0 = 5 \wedge y = x + 1$$

$$x := y \quad \exists x_0, y = x_0 + 1 \wedge x = y$$

$$\boxed{y = 5 = x = 5}$$

$$y = 5 = x = 5$$

$$8. SP [x := x+1; x := x+1, x \leq 10]$$

$$x_0 \leq 10 \wedge x = x_0 + 1, x \leq 11$$

$$x_0 \leq 11 \wedge x = x_0 + 1 \Rightarrow (x \leq 12)$$

$$9. SP [b := a+c, a := 0, a < c]$$

$$a_0 < c \wedge a = 0$$

$$a = 0 \wedge c > 0$$

$$b = a+c$$

$$b = c$$

$$a = 0 \wedge c > 0 \wedge b = c$$

$$10. SP [y := x-5; x := y+1; x \geq 5]$$

$$x = y+1 \wedge x \geq 5 \wedge y = x-5$$

$$x \geq 5 \wedge y = x-5$$

$$11. SP [if x \geq 0 \text{ then } x := x+2 \text{ else } x := x-2, T]$$

$$① x = x_0 + 2, x \geq 2$$

$$② x \geq 0 \rightarrow x < 0, x = x_0 - 2, x_0 = x + 2, x < -2$$

$$(x \geq 2) \vee (x < -2)$$

$$12. SP [if x > y \text{ then } z := x-y \text{ else } z := y-x, x \geq 0 \wedge y \geq 0]$$

$$① x > y, x \geq 0, y \geq 0, z = x-y$$

no further simplification

$$② y > x, x \geq 0, y \geq 0, z = y-x$$

$$(x > y \wedge x \geq 0 \wedge y \geq 0 \wedge z = x-y) \wedge (y > x \wedge x \geq 0 \wedge y \geq 0 \wedge z = y-x)$$

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13. SP $[x := x+1, y := y+1, x=y]$

$$\neg x_0 = y \wedge x = x_0 + 1$$

$$\therefore x = y + 1$$

$$\rightarrow y_0, x = y_0 + 1 \wedge y = y_0 + 1 \rightarrow x = y + 1$$

14. SP $[a := a-1, b := b-1, a < b]$

$$\neg a_0 < b, a = a_0 - 1 \wedge a < b - 1$$

$$\rightarrow b_0, a < b_0 - 1, b = b_0 - 1 \quad (a < b)$$

15. SP $[\text{if } x > 1 \text{ then } x := x-1 \text{ else } x := x+2, x \geq 1]$

$$\textcircled{1} x > 1, \neg x_0, x = x_0 - 1 \wedge x_0 \geq 1, \quad (x > 0)$$

$$\textcircled{2} x \leq 1, \neg x_0, x = x_0 + 2 \wedge x_0 \geq 2$$

contradiction.

16. SP $[\text{if } x > 10, x := 1 \text{ else } x := 2, x > 0]$

$$\textcircled{1} x > 10, x = 1$$

$$(x=1) \wedge (x=2)$$

$$\textcircled{2} x \leq 10, x = 2$$

①

②

17. SP $[x := x+1, \text{ if } x > 0 \text{ then } y := y+x, x \geq 0 \wedge y = 0]$

$$x = x + 1, x_0 \geq 0, y = 0, x = x_0 + 1 \therefore x \geq 1$$

$$x \geq 0, x \geq 1$$

$$y = y_0 + x \quad y = x$$

$$y = 0,$$

$$(x \geq 1 \wedge y = x)$$

18. SP [if $x \geq y$ $z := x - y$ else $z := y - x$; $x \geq 0 \wedge y \geq 0$]

① $x \geq y, x \geq 0, y \geq 0 \quad z = x - y$

② $x < y, x \geq 0, y \geq 0, z = y - x$

$(x \geq y \wedge x \geq 0 \wedge y \geq 0 \wedge z = x - y) \wedge (x < y \wedge x \geq 0 \wedge y \geq 0 \wedge z = y - x)$

①

②

19. SP [$z := x, x := y, y := z, x = y$]

$z := x, z = x$

$x := y, x = y$

$y := z, y = z$

$x = y = z$

20. SP [$y := x + 3; z := y \times 2; x := z - 1, x \geq 0$]

$x = 0, y := x + 3, y = 0 + 3 = y = 3$

$x = 0, z := y \times 2 = 3 \times 2 = 6 = z = 6$

$x = 0, x := z - 1 = 6 - 1 = 5 \geq x \geq 5$

$x = 5 \wedge y = 3 \wedge z = 6$